

Fourier Series



VIT-AP
UNIVERSITY

Dr. Aswathy R K

Department of Mathematics

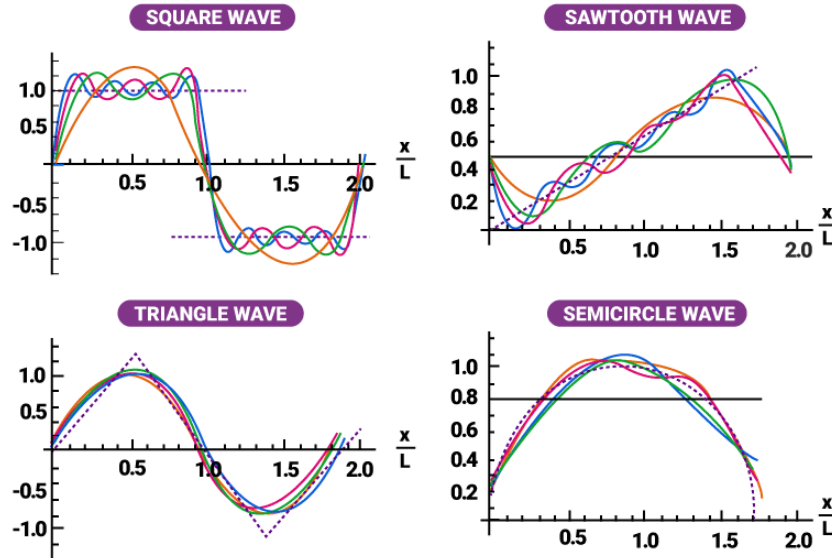
School of Advanced Sciences

Vellore Institute of Technology – Andhra Pradesh

Fourier Series

The theory of Fourier series and the Fourier transform is concerned with dividing a function into a superposition of sines and cosines, its components of various frequencies.

Fourier analysis is a mathematical technique which enables us to decompose an arbitrary function into a superposition of oscillations.



Periodic Functions

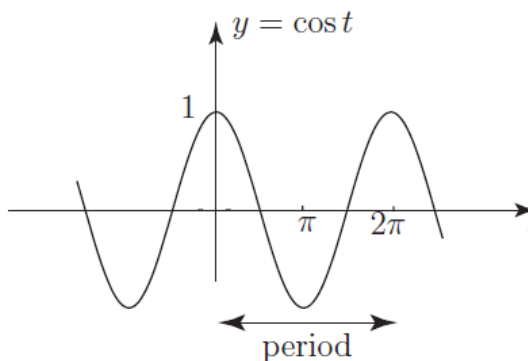
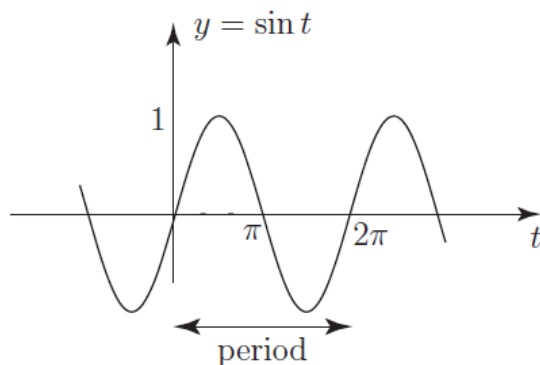
Definition (Even and Odd Function). A $f(x)$ is said to be even if $f(-x) = f(x)$

and odd if $f(-x) = -f(x)$.

Definition (Periodic Function). A $f(x)$ is said to be periodic with period T if

$$f(x + T) = f(x), \quad T > 0.$$

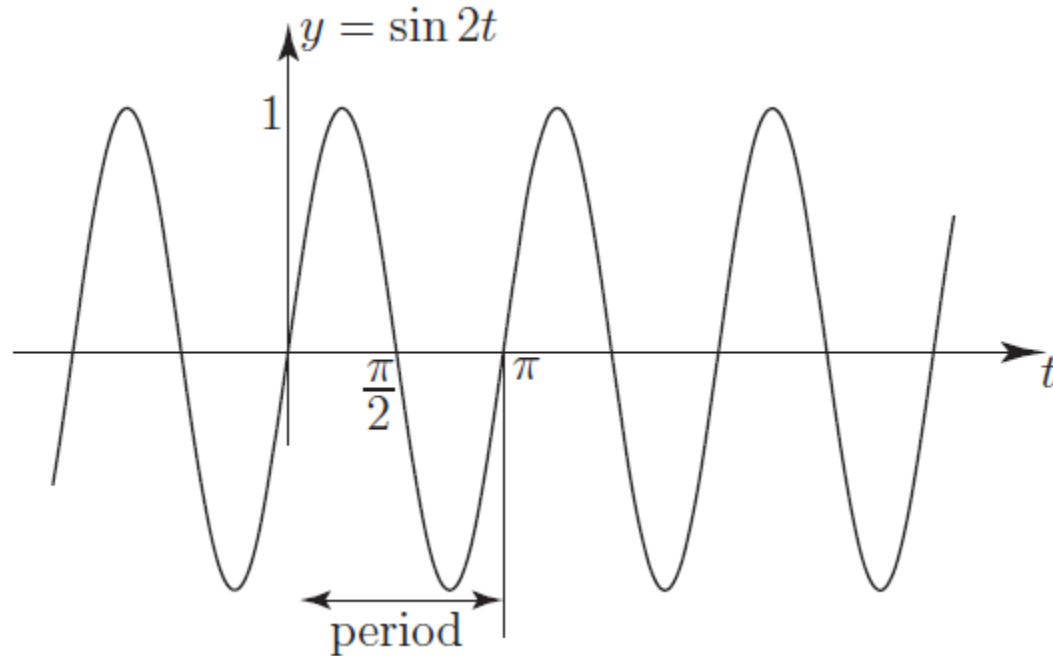
$$\sin(t + 2\pi) = \sin t \quad \text{and} \quad \cos(t + 2\pi) = \cos t$$



Periodic Functions

$$y = \sin 2t$$

$$\sin 2(t + \pi) = \sin(2t + 2\pi) = \sin 2t$$



$$\sin 5x = \sin (5x + 2\pi) = \sin 5 \left(x + \frac{2\pi}{5} \right), \text{Period} = \frac{2\pi}{5}$$

$$\cos 3x = \cos (3x + 2\pi) = \cos 3 \left(x + \frac{2\pi}{3} \right), \text{Period} = \frac{2\pi}{3}$$

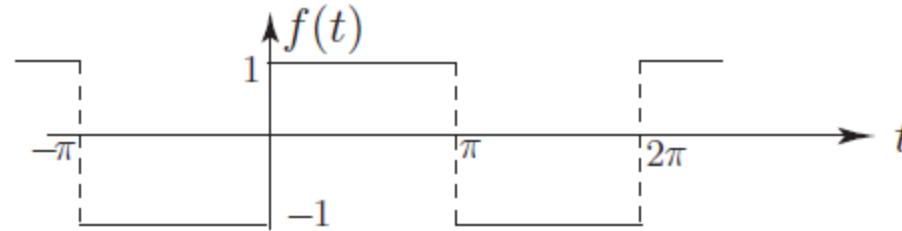
$$\begin{aligned} \cos \frac{2n\pi x}{k} &= \cos \left(\frac{2n\pi x}{k} + 2\pi \right) = \cos \frac{2n\pi}{k} \left(x + \frac{2\pi k}{2n\pi} \right) \\ &= \cos \frac{2n\pi}{k} \left(x + \frac{k}{n} \right), \text{Period} = \frac{k}{n} \end{aligned}$$

$$\tan 2x = \tan (2x + \pi) = \tan 2 \left(x + \frac{\pi}{2} \right), \text{Period} = \frac{\pi}{2}$$

Theorem *Any real valued function $f(x)$ can written as sum of even and odd function as:*

$$f(x) = \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2}$$

Square wave

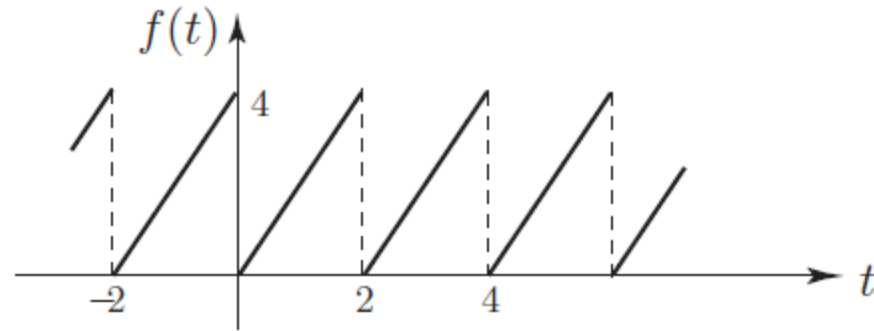


Analytically we can describe this function as follows:

$$f(t) = \begin{cases} -1 & -\pi < t < 0 \\ +1 & 0 < t < \pi \end{cases} \quad (\text{which gives the definition over one period})$$

$$f(t + 2\pi) = f(t) \quad (\text{which tells us that the function has period } 2\pi)$$

Saw-tooth wave

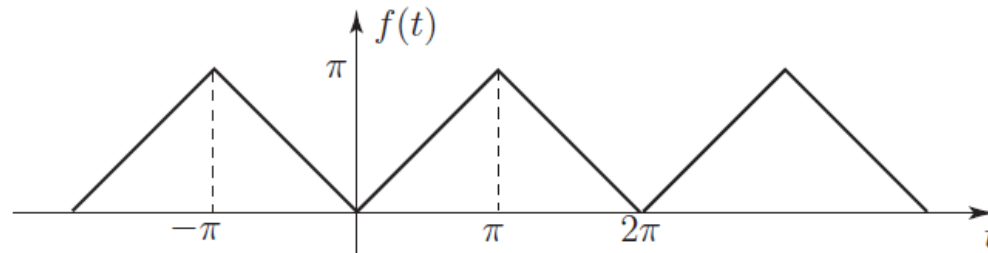


In this case we can describe the function as follows:

$$f(t) = 2t \qquad 0 < t < 2 \qquad f(t+2) = f(t)$$

Here the period is 2,

Triangular wave



Here we can conveniently define the function using $-\pi < t < \pi$ as the “basic period”:

$$f(t) = \begin{cases} -t & -\pi < t < 0 \\ t & 0 < t < \pi \end{cases}$$

or, more concisely,

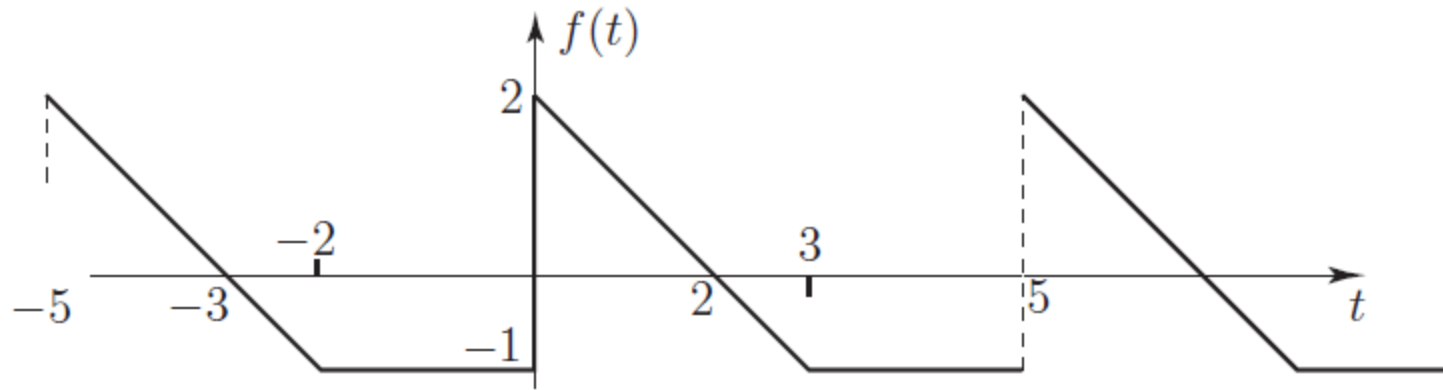
$$f(t) = |t| \quad -\pi < t < \pi$$

together with the usual statement on periodicity

$$f(t + 2\pi) = f(t).$$

$$f(t) = \begin{cases} 2 - t & 0 < t < 3 \\ -1 & 3 < t < 5 \end{cases}$$

$$f(t + 5) = f(t)$$

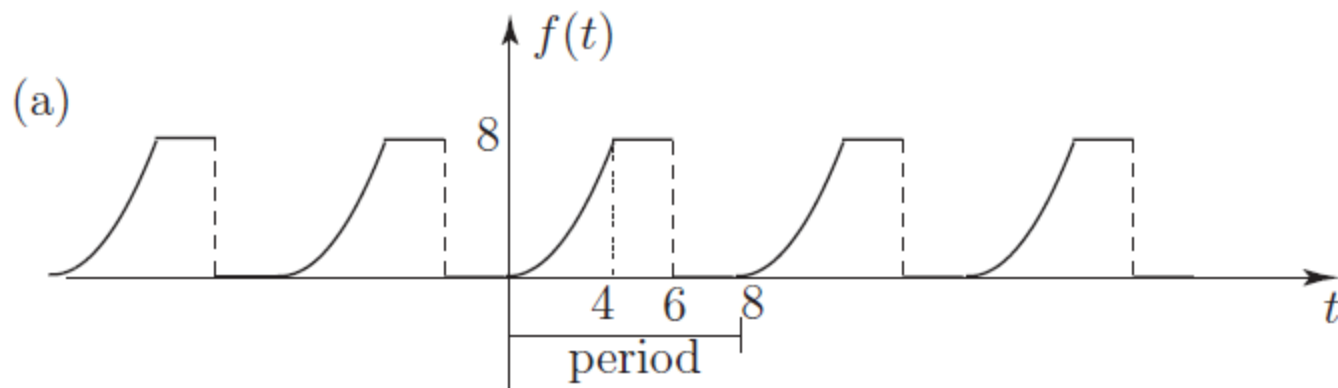


Periodic Functions

Sketch the graphs of the following periodic functions showing all relevant values:

$$(a) \quad f(t) = \begin{cases} t^2/2 & 0 < t < 4 \\ 8 & 4 < t < 6 \\ 0 & 6 < t < 8 \end{cases} \quad f(t+8) = f(t)$$

$$(b) \quad f(t) = 2t - t^2 \quad 0 < t < 2 \quad f(t+2) = f(t)$$

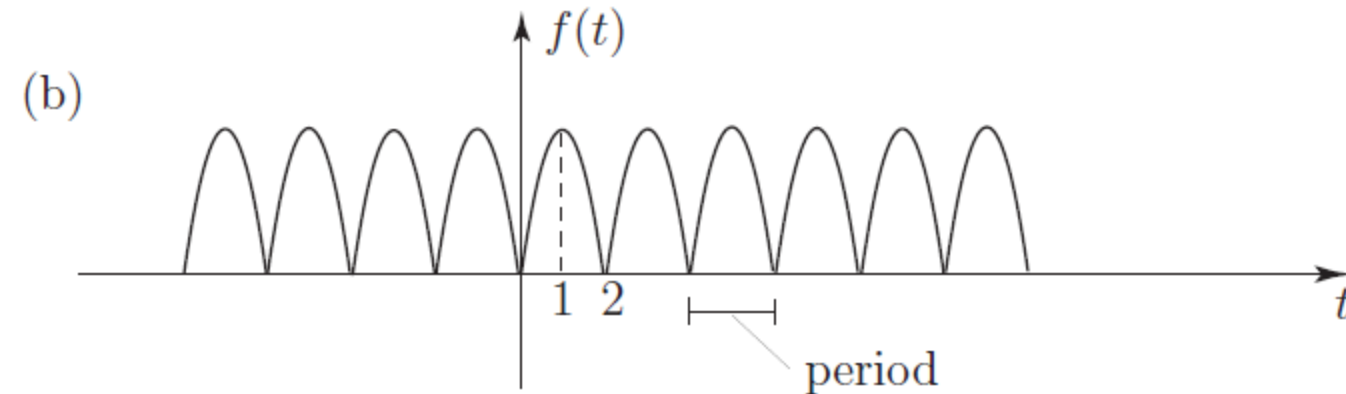


Periodic Functions

Sketch the graphs of the following periodic functions showing all relevant values:

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$$(b) \quad f(t) = 2t - t^2 \quad 0 < t < 2 \quad f(t+2) = f(t)$$



Let $f(x)$ be a real values function defined over some domain $[-l \ l]$ and periodic with period $2l$. The Fourier series of $f(x)$ is defined as:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$$

where

$$\underbrace{a_0 = \frac{1}{l} \int_{-l}^l f(x) dx \qquad a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx \qquad b_n = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx}_{\text{Euler's Formulas for Fourier Coefficients}}$$

Euler's Formulas for Fourier Coefficients

ORTHOGONALITY CONDITIONS FOR THE SINE AND COSINE FUNCTIONS

Notice that the Fourier coefficients are integrals. These are obtained by starting with the series, and employing the following properties called orthogonality conditions:

$$(a) \int_{-L}^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = 0 \text{ if } m \neq n \text{ and } L \text{ if } m = n$$

$$(b) \int_{-L}^L \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx = 0 \text{ if } m \neq n \text{ and } L \text{ if } m = n$$

$$(c) \int_{-L}^L \sin \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = 0. \text{ Where } m \text{ and } n \text{ can assume any positive integer values.}$$

find the Fourier coefficients of the function extended to be periodic of period 2π .

$$f(t) = \begin{cases} -\pi, & \text{for } -\pi < t < 0 \\ \pi, & \text{for } 0 < t < \pi \\ 0, & \text{for } t = 0, \pi, \end{cases}$$

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx$$

$$\frac{a_0}{2} = \text{average value of } f = 0,$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$a_m = \frac{1}{\pi} \left[\int_{-\pi}^{\pi} f(t) \cos mtdt \right] = \frac{1}{\pi} \left[\int_{-\pi}^0 -\pi \cos mtdt \right] + \frac{1}{\pi} \left[\int_0^{\pi} \pi \cos mtdt \right]$$

$$= \frac{1}{\pi} \frac{-\pi}{m} [\sin mt]_{-\pi}^0 + \frac{1}{\pi} \frac{\pi}{m} [\sin mt]_0^{\pi} = \dots = 0,$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$b_m = \frac{1}{\pi} \left[\int_{-\pi}^{\pi} f(t) \sin mtdt \right] = \frac{1}{\pi} \left[\int_{-\pi}^0 -\pi \sin mtdt \right] + \frac{1}{\pi} \left[\int_0^{\pi} \pi \sin mtdt \right]$$

$$= \frac{1}{\pi} \frac{\pi}{m} [\cos mt]_{-\pi}^0 + \frac{1}{\pi} \frac{-\pi}{m} [\cos mt]_0^{\pi} = \frac{2}{m} - \frac{2}{m} \cos m\pi$$

$$= \frac{2}{m} (1 - (-1)^m) = \begin{cases} \frac{4}{m}, & \text{if } m \text{ is odd,} \\ 0, & \text{if } m \text{ is even.} \end{cases}$$

Thus we find the Fourier series for f :

$$f(t) = 4 \sin t + \frac{4}{3} \sin 3t + \frac{4}{5} \sin 5t + \cdots$$

The trigonometric polynomials

$$\phi_1(t) = 4 \sin t, \quad \phi_3(t) = 4 \sin t + \frac{4}{3} \sin 3t \quad \phi_5(t) = 4 \sin t + \frac{4}{3} \sin 3t + \frac{4}{5} \sin 5t$$

are approximations to $f(t)$ which improve as the number of terms increases.

If we set $x = \pi/2$, $f(x) = \pi$,

$$\begin{aligned} \pi &= 4 \sin(\pi/2) + \frac{4}{3} \sin(3\pi/2) + \frac{4}{5} \sin(5\pi/2) + \cdots \\ &= 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots\right). \end{aligned}$$

DIRICHLET CONDITIONS

Suppose that

- (1) $f(x)$ is defined except possibly at a finite number of points in $(-L, L)$
- (2) $f(x)$ is periodic outside $(-L, L)$ with period $2L$
- (3) $f(x)$ and $f'(x)$ are piecewise continuous in $(-L, L)$.

Then the series with Fourier coefficients converges to

(a) $f(x)$ if x is a point of continuity

(b) $\frac{f(x+0) + f(x-0)}{2}$ if x is a point of discontinuity

Here $f(x+0)$ and $f(x-0)$ are the right- and left-hand limits of $f(x)$ at x and represent $\lim_{\epsilon \rightarrow 0+} f(x+\epsilon)$ and $\lim_{\epsilon \rightarrow 0+} f(x-\epsilon)$, respectively.

USEFUL INTEGRALS

The following integrals are useful in Fourier Series.

$$(i) \int_0^{2\pi} \sin nx \, dx = 0$$

$$(ii) \int_0^{2\pi} \cos nx \, dx = 0$$

$$(iii) \int_0^{2\pi} \sin^2 nx \, dx = \pi$$

$$(iv) \int_0^{2\pi} \cos^2 nx \, dx = \pi$$

$$(v) \int_0^{2\pi} \sin nx \cdot \sin mx \, dx = 0$$

$$(vi) \int_0^{2\pi} \cos nx \cos mx \, dx = 0$$

$$(vii) \int_0^{2\pi} \sin nx \cdot \cos mx \, dx = 0$$

$$(viii) \int_0^{2\pi} \sin nx \cdot \cos nx \, dx = 0$$

$$(ix) [uv]_1 = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \dots$$

where $v_1 = \int v \, dx$, $v_2 = \int v_1 \, dx$ and so on $u' = \frac{du}{dx}$, $u'' = \frac{d^2u}{dx^2}$ and so on and

$$(x) \sin n\pi = 0, \cos n\pi = (-1)^n \text{ where } n \in I$$

if m and n are positive integers,

$$\int_{-\pi}^{\pi} \cos nt \cos mt dt = \begin{cases} \pi, & \text{for } m = n, \\ 0, & m \neq n \end{cases},$$

$$\int_{-\pi}^{\pi} \sin nt \sin mt dt = \begin{cases} \pi & \text{for } m = n \\ 0 & \text{for } m \neq n \end{cases}$$

$$\int_{-\pi}^{\pi} \sin nt \cos mt dt = 0.$$

A function $f(x)$ can be expressed as a Fourier series in $(0, 2L)$ as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right]$$

$$\text{Where } a_0 = \frac{1}{L} \int_0^{2L} f(x) dx$$

$$a_n = \frac{1}{L} \int_0^{2L} f(x) \cos \frac{n\pi x}{L} dx$$

$$b_n = \frac{1}{L} \int_0^{2L} f(x) \sin \frac{n\pi x}{L} dx$$

Example 1

Obtain the Fourier series of the following function defined in $(0, 2\pi)$.

$$f(x) = \begin{cases} x, & 0 < x < \pi \\ \pi, & \pi < x < 2\pi, \end{cases} \quad \text{and has period } 2\pi$$

Solution.

STEP ONE

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} f(x) dx + \frac{1}{\pi} \int_{\pi}^{2\pi} f(x) dx \\ &= \frac{1}{\pi} \int_0^{\pi} x dx + \frac{1}{\pi} \int_{\pi}^{2\pi} \pi \cdot dx = \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} + \frac{\pi}{\pi} \left[x \right]_{\pi}^{2\pi} = \frac{\pi}{2} + \pi \\ &\text{i.e. } a_0 = \frac{3\pi}{2}. \end{aligned}$$

STEP TWO

$$\begin{aligned}a_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx \, dx \\&= \frac{1}{\pi} \int_0^{\pi} x \cos nx \, dx + \frac{1}{\pi} \int_{\pi}^{2\pi} \pi \cdot \cos nx \, dx \\&= \frac{1}{\pi} \left[\frac{1}{n} \left(\pi \sin n\pi - 0 \cdot \sin n0 \right) - \left[\frac{-\cos nx}{n^2} \right]_0^{\pi} \right] \\&\quad + \frac{1}{n} (\sin n2\pi - \sin n\pi)\end{aligned}$$

$$a_n = \frac{1}{\pi} \left[\frac{1}{n} (0 - 0) + \left(\frac{\cos n\pi}{n^2} - \frac{\cos 0}{n^2} \right) \right] + \frac{1}{n} (0 - 0)$$

$$= \frac{1}{n^2 \pi} (\cos n\pi - 1),$$

$$a_n = \begin{cases} -\frac{2}{n^2 \pi} & , n \text{ odd} \\ 0 & , n \text{ even.} \end{cases}$$

STEP THREE

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx \\
 &= \frac{1}{\pi} \int_0^{\pi} x \sin nx \, dx + \frac{1}{\pi} \int_{\pi}^{2\pi} \pi \cdot \sin nx \, dx \\
 &= \frac{1}{\pi} \left[\left(\frac{-\pi \cos n\pi}{n} + 0 \right) + \left[\frac{\sin nx}{n^2} \right]_0^{\pi} \right] - \frac{1}{n} (\cos 2n\pi - \cos n\pi) \\
 &= \frac{1}{\pi} \left[\frac{-\pi(-1)^n}{n} + \left(\frac{\sin n\pi - \sin 0}{n^2} \right) \right] - \frac{1}{n} (1 - (-1)^n) \\
 &= -\frac{1}{n}(-1)^n + 0 - \frac{1}{n}(1 - (-1)^n) = -\frac{1}{n}
 \end{aligned}$$

We now have

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$$

$$\text{where } a_0 = \frac{3\pi}{2}, \quad a_n = \begin{cases} 0 & , n \text{ even} \\ -\frac{2}{n^2\pi} & , n \text{ odd} \end{cases}, \quad b_n = -\frac{1}{n}$$

Fourier Series

Example 2. Given that $f(x) = x + x^2$ for $-\pi < x < \pi$, find the Fourier expression of $f(x)$.

Deduce that $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$

Solution. Let $x + x^2 = \frac{a_0}{2} + a_1 \cos x + a_2 \cos 2x + \dots + b_1 \sin x + b_2 \sin 2x + \dots$

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) dx \\ &= \frac{1}{\pi} \left[\frac{x^2}{2} + \frac{x^3}{3} \right]_{-\pi}^{\pi} = \frac{1}{\pi} \left[\frac{\pi^2}{2} + \frac{\pi^3}{3} - \frac{\pi^2}{2} + \frac{\pi^3}{3} \right] = \frac{2\pi^2}{3} \end{aligned}$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \int_{-\pi}^{\pi} (x + x^2) \cos nx dx \\ &= \frac{1}{\pi} \left[(x + x^2) \frac{\sin nx}{n} - (2x + 1) \frac{(-\cos nx)}{n^2} + (2) \left(-\frac{\sin nx}{n^3} \right) \right]_{-\pi}^{\pi} \\ &= \frac{1}{\pi} \left[(2\pi + 1) \frac{\cos n\pi}{n^2} - (-2\pi + 1) \frac{\cos(-n\pi)}{n^2} \right] = \frac{1}{\pi} \left[4\pi \frac{\cos n\pi}{n^2} \right] = \frac{4(-1)^n}{n^2} \end{aligned}$$

Fourier Series

$$\begin{aligned}b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) \sin nx \, dx \\&= \frac{1}{\pi} \left[(x + x^2) \left(-\frac{\cos nx}{n} \right) - (2x + 1) \left(\frac{-\sin nx}{n^2} \right) + 2 \frac{\cos nx}{n^3} \right]_{-\pi}^{\pi} \\&= \frac{1}{\pi} \left[-(\pi + \pi^2) \frac{\cos n\pi}{n} + 2 \frac{\cos n\pi}{n^3} + (-\pi + \pi^2) \frac{\cos n\pi}{n} - 2 \frac{\cos n\pi}{n^3} \right] \\&= \frac{1}{\pi} \left[-\frac{2\pi}{n} \cos n\pi \right] = -\frac{2}{n} (-1)^n\end{aligned}$$

Substituting the values of a_0 , a_n , b_n ,

$$\begin{aligned}x + x^2 &= \frac{\pi^2}{3} + 4 \left[-\cos x + \frac{1}{2^2} \cos 2x - \frac{1}{3^2} \cos 3x + \dots \right] \\&\quad - 2 \left[-\sin x + \frac{1}{2} \sin 2x - \frac{1}{3} \sin 3x + \dots \right]\end{aligned}$$

EXERCISE

1. Find a Fourier series to represent, $f(x) = \pi - x$ for $0 < x < 2\pi$.

$$\text{Ans. } 2 \left[\sin x + \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x + \dots + \frac{1}{n} \sin nx + \dots \right]$$

2. Find a Fourier series to represent $x - x^2$ from $x = -\pi$ to π and show that

$$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$$

$$\text{Ans. } -\frac{\pi^2}{3} + 4 \left[\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \frac{\cos 4x}{4^2} + \dots \right] + 2 \left[\frac{\sin x}{1} - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \frac{\sin 4x}{4} + \dots \right]$$

3. Find a Fourier series to represent the function $f(x) = e^x$, for $-\pi < x < \pi$ and hence derive a series for $\frac{\pi}{\sinh \pi}$.

$$\text{Ans. } \frac{2 \sinh \pi}{\pi} \left[\frac{1}{2} - \frac{1}{1^2 + 1} \cos x + \frac{1}{2^2 + 1} \cos 2x - \frac{1}{3^2 + 1} \cos 3x + \dots \right. \\ \left. + \frac{1}{1^2 + 1} \sin x - \frac{2}{2^2 + 1} \sin 2x - \frac{3}{3^2 + 1} \sin 3x + \dots \right], \quad \frac{\pi}{\sinh \pi} = 1 + 2 \left[-\frac{1}{2} + \frac{1}{5} - \frac{1}{10} + \dots \right]$$

4. Obtain the Fourier series for $f(x) = e^{-x}$ in the interval $0 \leq x < 2\pi$.

$$\text{Ans. } \frac{1 - e^{-2\pi}}{\pi} \left[\frac{1}{2} + \frac{1}{2} \cos x + \frac{1}{5} \cos 2x + \frac{1}{10} \cos 3x + \frac{1}{2} \sin x + \frac{2}{5} \sin 2x + \frac{3}{10} \sin 3x + \dots \right]$$

EXERCISE

1. Find the Fourier series of the function

$$f(x) = \begin{cases} -1 & \text{for } -\pi < x < 0 \\ 1 & \text{for } 0 < x < \pi \end{cases}$$

where $f(x + 2\pi) = f(x)$.

$$\text{Ans. } \frac{4}{\pi} \left[\frac{1}{1} \sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \frac{1}{7} \sin 7x + \dots \right]$$

2. Find the Fourier series of the function

$$f(x) = \begin{cases} 0 & \text{for } -\pi \leq x \leq 0 \\ 1 & \text{for } 0 < x < \frac{\pi}{2} \\ 0 & \text{for } \frac{\pi}{2} \leq x \leq \pi \end{cases}$$

$$\text{Ans. } \frac{1}{4} + \frac{1}{\pi} \left[\frac{\cos x}{1} - \frac{\cos 3x}{3} + \frac{\cos 5x}{5} + \dots + \frac{\sin x}{1} + \frac{\sin 2x}{2} + \frac{\sin 3x}{3} + \dots \right]$$

3. Obtain a Fourier series to represent the following periodic function

$f(x) = 0$ when $0 < x < \pi$

$f(x) = 1$ when $\pi < x < 2\pi$

$$\text{Ans. } \frac{1}{2} - \frac{2}{\pi} \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right)$$

If a function $f(x)$ is an even function in $(-L, L)$ then its Fourier series expansion contains

cosine terms only, i.e. $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{L} \right]$

$$a_0 = \frac{2}{L} \int_0^L f(x) dx$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$$

If a function $f(x)$ is an odd function in $(-L, L)$ then its Fourier series expansion contains

sine terms only, i.e. $f(x) = \sum_{n=1}^{\infty} \left[b_n \sin \frac{n\pi x}{L} \right]$

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

If f is defined over the half interval $0 \leq x \leq L$. We can write a Fourier cosine series and Fourier Sine series for f on $[0, L]$.

Cosine Series

Suppose $f(x)$ is defined for $0 \leq x \leq L$. To get pure Cosine series on this interval we defined

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

$$\text{where } a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

If f is defined over the half interval $0 \leq x \leq L$. We can write a Fourier cosine series and Fourier Sine series for f on $[0, L]$.

Sine Series

Let $f(x)$ is defined over $[0, L]$. We can defined Sine series that contains only Sine terms as

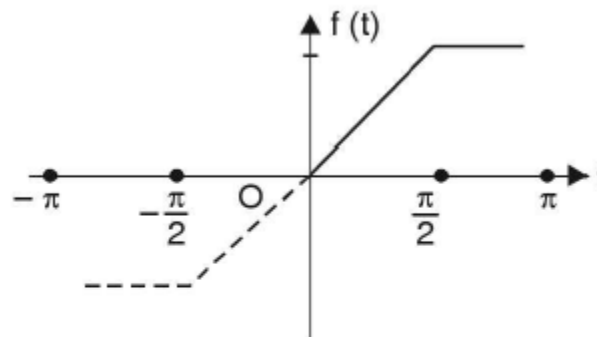
$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$\text{where } b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

Fourier Series – Half Range Series

Example 1. Represent the following function by a Fourier sine series :

$$f(t) = \begin{cases} t, & 0 < t \leq \frac{\pi}{2} \\ \frac{\pi}{2}, & \frac{\pi}{2} < t \leq \pi \end{cases}$$



Solution. $b_n = \frac{2}{\pi} \int_0^{\pi} f(t) \sin nt \, dt$

$$= \frac{2}{\pi} \int_0^{\pi/2} t \sin nt \, dt + \frac{2}{\pi} \int_{\pi/2}^{\pi} \frac{\pi}{2} \sin nt \, dt$$

$$= \frac{2}{\pi} \left[t \left(-\frac{\cos nt}{n} \right) - (1) \left(-\frac{\sin nt}{n^2} \right) \right]_0^{\pi/2} + \frac{2\pi}{\pi^2} \left[-\frac{\cos nt}{n} \right]_{\pi/2}^{\pi}$$

$$= \frac{2}{\pi} \left[-\frac{\pi}{2} \frac{\cos \frac{n\pi}{2}}{n} + \frac{\sin \frac{n\pi}{2}}{n^2} \right] + \left[-\frac{\cos n\pi}{n} + \frac{\cos \frac{n\pi}{2}}{n} \right]$$

$$b_1 = \frac{2}{\pi} \left[-\frac{\pi}{2} \cos \frac{\pi}{2} + \sin \frac{\pi}{2} \right] + \left[-\cos \pi + \cos \frac{\pi}{2} \right] = \frac{2}{\pi} [0 + 1] + [1] = \frac{2}{\pi} + 1$$

$$\begin{aligned} b_2 &= \frac{2}{\pi} \left[-\frac{\pi}{2} \frac{\cos \pi}{2} + \frac{\sin \pi}{2^2} \right] + \left[-\frac{\cos 2\pi}{2} + \frac{\cos \pi}{2} \right] = \frac{2}{\pi} \left[-\frac{\pi}{2} \frac{(-1)}{2} + 0 \right] + \left[-\frac{1}{2} - \frac{1}{2} \right] \\ &= \frac{2}{\pi} \left[\frac{\pi}{4} \right] - 1 = \frac{1}{2} - 1 = -\frac{1}{2} \end{aligned}$$

$$\begin{aligned} b_3 &= \frac{2}{\pi} \left[-\frac{\pi}{2} \frac{\cos \frac{3\pi}{2}}{3} + \frac{\sin \frac{3\pi}{2}}{3^2} \right] + \left[-\frac{\cos 3\pi}{3} + \frac{\cos \frac{3\pi}{2}}{3} \right] \\ &= \frac{2}{\pi} \left[-\frac{\pi}{2} (0) - \frac{1}{9} \right] + \left[\frac{1}{3} + 0 \right] = -\frac{2}{9\pi} + \frac{1}{3} \end{aligned}$$

$$f(t) = \left(\frac{2}{\pi} + 1 \right) \sin t - \frac{1}{2} \sin 2t + \left(-\frac{2}{9\pi} + \frac{1}{3} \right) \sin 3t + \dots$$

Fourier Series – Half Range Series

Example 1. Find the Fourier sine series for the function

$$f(x) = e^{ax} \text{ for } 0 < x < \pi$$

where a is constant.

Solution.

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^{\pi} e^{ax} \sin nx \, dx \quad \left[\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx] \right] \\ &= \frac{2}{\pi} \left[\frac{e^{ax}}{a^2 + n^2} (a \sin nx - n \cos nx) \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[\frac{e^{a\pi}}{a^2 + n^2} (a \sin n\pi - n \cos n\pi) + \frac{n}{a^2 + n^2} \right] \\ &= \frac{2}{\pi} \frac{n}{a^2 + n^2} [-(-1)^n e^{a\pi} + 1] = \frac{2n}{(a^2 + n^2)\pi} [1 - (-1)^n e^{a\pi}] \end{aligned}$$

$$b_1 = \frac{2(1 + e^{a\pi})}{(a^2 + 1^2)\pi}, \quad b_2 = \frac{2 \cdot 2 \cdot (1 - e^{a\pi})}{(a^2 + 2^2)\pi}$$
$$e^{ax} = \frac{2}{\pi} \left[\frac{1 + e^{a\pi}}{a^2 + 1^2} \sin x + \frac{2(1 - e^{a\pi})}{a^2 + 2^2} \sin 2x + \dots \right]$$

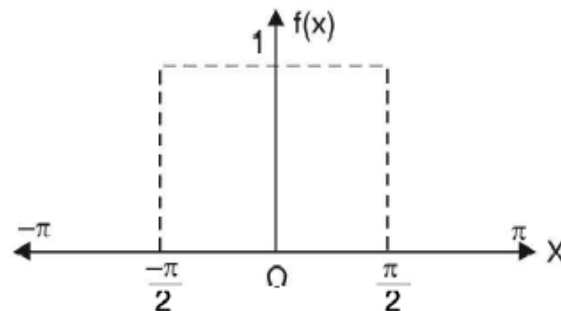
Fourier Series – Half Range Series

EXERCISE

1. Find the Fourier cosine series for the function

$$f(x) = \begin{cases} 1 & \text{for } 0 < x < \frac{\pi}{2} \\ 0 & \text{for } \frac{\pi}{2} < x < \pi. \end{cases}$$

Ans. $\frac{1}{2} + \frac{2}{\pi} \left[\cos x - \frac{1}{3} \cos 3x + \frac{1}{5} \cos 5x - \dots \right]$



2. Find a series of cosine of multiples of x which will represent $f(x)$ in $(0, \pi)$ where

$$f(x) = \begin{cases} 0 & \text{for } 0 < x < \frac{\pi}{2} \\ \frac{\pi}{2} & \text{for } \frac{\pi}{2} < x < \pi \end{cases}$$

Deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \infty = \frac{\pi}{4}$

Ans. $\frac{\pi}{4} - \cos x + \frac{1}{3} \cos 3x - \frac{1}{5} \cos 5x + \dots$

3. Express $f(x) = x$ as a sine series in $0 < x < \pi$.

Ans. $2 \left[\sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x - \dots \right]$

4. Find the cosine series for $f(x) = \pi - x$ in the interval $0 < x < \pi$.

Ans. $\frac{\pi}{2} + \frac{4}{\pi} \left(\cos x + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right)$

Theorem (Parseval's Theorem). *Let f be continuous function on $[-L, L]$ and let f' be piecewise continuous. then the Fourier coefficients of f on $[-L, L]$ satisfy*

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) = \frac{1}{L} \int_{-L}^L f(x)^2 dx.$$

and we can define Parseval's Identity for half range series as

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} a_n^2 = \frac{2}{L} \int_0^L f(x)^2 dx. \quad \sum_{n=1}^{\infty} b_n^2 = \frac{2}{L} \int_0^L f(x)^2 dx.$$

Parseval's identity in $(0, 2L)$

$$\int_0^{2L} |f(x)|^2 dx = L \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} [a_n^2 + b_n^2] \right]$$

Parseval's identity in $(-L, L)$

$$\int_{-L}^L |f(x)|^2 dx = L \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} [a_n^2 + b_n^2] \right]$$

Parseval's identity in $(0, L)$ provided half range cosine series

$$\int_0^L |f(x)|^2 dx = \frac{L}{2} \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} [a_n^2] \right]$$

Parseval's identity in $(0, L)$ provided half range sine series

$$\int_0^L |f(x)|^2 dx = \frac{L}{2} \left[\sum_{n=1}^{\infty} [b_n^2] \right]$$

Definition: The **root mean square** (r.m.s) value of a function $y = f(x)$ over an interval from $x = a$ to $x = b$ is defined as:

$$\bar{y} = \sqrt{\frac{\int_a^b y^2 dx}{b - a}}$$

If we take the interval as $(c, c + 2l)$,

Then

$$\bar{y}^2 = \frac{1}{2l} \int_c^{c+2l} y^2 dx$$

Fourier Series – Parseval's Identity

Example . By using the series for $f(x) = 1$ in $0 < x < \pi$ show that

$$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$$

Solution. Sine series is $f(x) = \sum b_n \sin nx$

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^\pi f(x) \sin nx \, dx \\ &= \frac{2}{\pi} \int_0^\pi (1) \sin nx \, dx = \frac{2}{\pi} \left(\frac{-\cos nx}{n} \right)_0^\pi = \frac{-2}{n\pi} [\cos n\pi - 1] = \frac{-2}{n\pi} [(-1)^n - 1] \end{aligned}$$

$$= \frac{4}{n\pi} \text{ if } n \text{ is odd} = 0 \text{ if } n \text{ is even}$$

Then the sine series is

$$1 = \frac{4}{\pi} \sin x + \frac{4}{3\pi} \sin 3x + \frac{4}{5\pi} \sin 5x + \frac{4}{7\pi} \sin 7x + \dots$$

Fourier Series – Parseval's Identity

$$\int_0^c [f(x)]^2 dx = \frac{c}{2} [b_1^2 + b_2^2 + b_3^2 + b_4^2 + b_5^2 + \dots]$$

$$\int_0^\pi (1)^2 dx = \frac{\pi}{2} \left[\left(\frac{4}{\pi} \right)^2 + \left(\frac{4}{3\pi} \right)^2 + \left(\frac{4}{5\pi} \right)^2 + \left(\frac{4}{7\pi} \right)^2 + \dots \right]$$

$$[x]_0^\pi = \left(\frac{\pi}{2} \right) \left(\frac{16}{\pi^2} \right) \left[1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots \right]$$

$$\pi = \frac{\pi}{2} \left(\frac{16}{\pi^2} \right) \left[1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots \right]$$

$$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$$

Fourier Series – Parseval's Identity

Example . If $f(x) = \begin{cases} \pi x & 0 < x < 1 \\ \pi(2-x), & 1 < x < 2 \end{cases}$

using half range cosine series, show that

$$\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots = \frac{\pi^4}{96}$$

Solution. Half range consine series is

$$f(x) = \frac{a_0}{2} + \sum a_n \cos \frac{n\pi x}{c}$$

$$\text{where } a_0 = \frac{2}{c} \int_0^c f(x) dx = \frac{2}{2} \left[\int_0^1 \pi x dx + \int_1^2 \pi (2-x) dx \right]$$

$$= \pi \left(\frac{x^2}{2} \right)_0^1 + \pi \left(2x - \frac{x^2}{2} \right)_1^2 = \frac{\pi}{2} + \pi \left[(4-2) - \left(2 - \frac{1}{2} \right) \right] = \pi$$

$$\begin{aligned} a_n &= \frac{2}{c} \int_0^c f(x) \cos \frac{n\pi x}{c} dx \\ &= \frac{2}{2} \left[\int_0^1 \pi x \cos \frac{n\pi x}{2} dx + \int_1^2 \pi (2-x) \cos \frac{n\pi x}{2} dx \right] \end{aligned}$$

Fourier Series – Parseval's Identity

$$\begin{aligned}
 &= \pi \left[\frac{x \frac{\sin n\pi x}{2}}{\frac{n\pi}{2}} - \left(\frac{-\cos \frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}} \right) \right]_0^1 + \pi \left[(2-x) \frac{\sin \frac{n\pi x}{2}}{\frac{n\pi}{2}} - (-1) \left(\frac{-\cos \frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}} \right) \right]_1^2 \\
 &= \pi \left[\frac{2}{n\pi} \sin \frac{n\pi}{2} + \frac{4}{n^2\pi^2} \cos \frac{n\pi}{2} - \frac{4}{n^2\pi^2} \right] + \pi \left[0 - \frac{4}{n^2\pi^2} \cos n\pi - \frac{2}{n\pi} \sin \frac{n\pi}{2} + \frac{4}{n^2\pi^2} \cos \frac{n\pi}{2} \right] \\
 &= \pi \left[\frac{8}{n^2\pi^2} \cos \frac{n\pi}{2} - \frac{4}{n^2\pi^2} - \frac{4}{n^2\pi^2} \cos n\pi \right] = \frac{4}{n^2\pi} \left[2 \cos \frac{n\pi}{2} - 1 - \cos n\pi \right]
 \end{aligned}$$

$$a_1 = 0, a_2 = \frac{-4}{\pi}, a_3 = 0, a_4 = 0, a_5 = 0, a_6 = \frac{-4}{9\pi} \dots$$

$$\int_0^c [f(x)]^2 dx = \frac{c}{2} \left[\frac{a_0^2}{2} + a_1^2 + a_2^2 + a_3^2 + \dots \right]$$

$$\int_0^1 (\pi x)^2 dx + \int_1^2 \pi^2 (2-x)^2 dx = \frac{2}{2} \left[\frac{\pi^2}{2} + \frac{16}{\pi^2} + \frac{16}{81\pi^2} + \dots \right]$$

Fourier Series – Parseval's Identity

$$\begin{aligned}\pi^2 \left[\frac{x^3}{3} \right]_0^1 - \pi^2 \left[\frac{(2-x)^3}{3} \right]_1^2 &= \frac{\pi^2}{2} + \frac{16}{\pi^2} + \frac{16}{81\pi^2} + \dots \\ \frac{\pi^2}{3} - \pi^2 \left(0 - \frac{1}{3} \right) &= \frac{\pi^2}{2} + \frac{16}{\pi^2} \left[1 + \frac{1}{81} + \dots \right] \\ \frac{2\pi^2}{3} - \frac{\pi^2}{2} &= \frac{16}{\pi^2} \left[1 + \frac{1}{3^4} + \frac{1}{5^4} + \dots \right] \\ \frac{\pi^2}{6} &= \frac{16}{\pi^2} \left[1 + \frac{1}{3^4} + \frac{1}{5^4} + \dots \right] \\ \frac{\pi^4}{96} &= 1 + \frac{1}{3^4} + \frac{1}{5^4} + \dots\end{aligned}$$

EXERCISE

1. Prove that in $0 < x < c$,

$$x = \frac{c}{2} - \frac{4c}{\pi^2} \left(\cos \frac{\pi x}{c} + \frac{1}{3^2} \cos \frac{3\pi x}{c} + \frac{1}{5^2} \cos \frac{5\pi x}{c} + \dots \right) \text{ and deduce that}$$

$$(i) \quad \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots = \frac{\pi^4}{96}$$

$$(ii) \quad \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \dots = \frac{\pi^4}{90}$$

Thank You...